

Senior Project Proposal

SARA (Smart Assistive Reading Aid): An Offline Smart Glass for Visually Impaired People that Reads Bangla and English Text, Detects Objects, and Recognises and Checks Bangladeshi Taka Notes

Prepared by
MD Abdun Fattah Lam
ID: 2220014
Mukut Protim Memo
ID: 2020086
Tanjila Hossain Tapti
ID: 2130561

Spring 2026

Supervisor
Mohammad Shidujaman, PhD
Assistant Professor
Department of Computer Science and Engineering
School of Engineering, Technology and Sciences
Independent University, Bangladesh

1. Introduction

A blind person in Bangladesh who buys medicine, hands over a banknote and receives change has to ask someone what the label says, which note came back, and whether that note is real. The World Health Organization estimates that at least 2.2 billion people live with some form of vision impairment, and a large share of them live in low- and middle-income countries [1]. For these people, reading a sign, recognising an object and handling cash are daily tasks that depend on the help of others.

Camera-based assistive devices can take over part of this help. Commercial products such as OrCam MyEye [2] show that the idea works, but they are expensive, are built mainly for English, and do not know Bangladeshi Taka. Low-cost research prototypes built on Raspberry Pi boards usually read English only and speak through an online service [3, 4]. We propose *SARA (Smart Assistive Reading Aid)*, a wearable smart glass that speaks Bangla, reads Bangla and English text, names everyday objects, recognises Taka notes and gives careful advice about counterfeit notes, and that does its main work on the device without the internet.

2. Problem Statement

A blind or low-vision person in Bangladesh has no affordable wearable device that reads Bangla text aloud, tells which Taka note is in the hand, and gives trustworthy advice on whether that note may be counterfeit, without a phone, a screen or an internet connection. Three things make this hard:

- **Language and currency.** Existing devices are designed for English and for other currencies. Bangla script, with its conjuncts and vowel signs, is harder for open OCR engines than English.
- **Connectivity and cost.** Many systems send pictures to a server or need a paid service. This fails where the connection is poor and adds a cost that many users cannot carry.
- **Trust.** A blind user cannot check the device's answer by sight. A wrong denomination or a wrong counterfeit verdict can cost money or lead to a false accusation, so the device must say when it is not sure.

3. Objectives

The aim is to design, build and evaluate a working prototype of SARA. The specific objectives are:

1 To build a wearable prototype on a Raspberry Pi 5 with a camera, operated only through physical buttons and speech.

2 To read printed Bangla and English text aloud using on-device OCR and offline text-to-speech.

3 To announce everyday objects in front of the user in Bangla.

4 To recognise all nine denominations of Bangladeshi Taka (2, 5, 10, 20, 50, 100, 200, 500 and 1000) in ordinary camera pictures, using training data that needs no manual labelling.

5 To study a camera-based check for counterfeit 500 and 1000 Taka notes, and to design spoken feedback that is safe for a user who cannot verify it.

6 To measure the accuracy of every function on held-out data, and its speed, memory use and temperature on the Raspberry Pi 5.

7 To try the prototype with visually impaired users and collect their feedback.

4. Scope

In scope: text reading (Bangla and English), object announcement, Taka denomination recognition, a counterfeit check for 500 and 1000 Taka notes, Bangla and English speech output, a five-button interface, a 3D-printed spectacle frame, and an evaluation of each function. An optional mode that describes a scene through an online vision model may be added; it will need the internet and will not be required by the other modes.

Out of scope: navigation and obstacle distance, GPS, handwriting recognition, and voice-command control. The glass is an aid. It will not replace a white cane, a guide, or the counterfeit-detection equipment of a bank.

5. Related Work and Gap

Reading aids for blind people have a long history, from devices that turned print into vibration to finger-worn cameras and smart glasses [2, 5]. Raspberry Pi readers combine a camera, an OCR engine and a speech engine at low cost, but mostly for English and with online speech [3, 4]. Recent glasses add scene description and face recognition by sending pictures to a remote server [6]. For Bangladeshi currency, detectors have been trained on a few thousand hand-labelled images [7], and a public dataset of genuine and counterfeit notes with six views per note is available [8]. Fast object detectors [9] and multilingual OCR engines [10] make on-device operation possible.

What is missing is the combination: one wearable device that reads Bangla, recognises Taka, speaks Bangla and works offline, together with an honest measurement of how well each part works on the target hardware and with its users. This project aims to fill that gap.

6. Proposed Methodology

The work will follow four phases: requirement analysis, data and model development, system integration, and evaluation. Each function will be a separate mode that the user selects with one button.

- **Text reading.** EasyOCR [10] with Bangla and English recognisers. The capture button stores the recognised text and a separate button speaks it, so the user holds the camera still only for a moment. We will test pre-processing, region selection and post-processing on a test set built for the project.
- **Object announcement.** A COCO-pretrained YOLOv8 detector [9] scans the scene at a fixed interval and speaks the most confident objects in Bangla, with a waiting time per object so that it does not repeat itself.
- **Currency recognition.** Public note images [11] will be pasted onto real photographs with random rotation, scale, lighting and blur, so that bounding-box labels are produced automatically. A YOLOv8 detector will be trained on these images and tested on photographs from other sources.
- **Counterfeit check.** A multi-view classifier will be trained on the JaalTaka dataset [8]. Because a blind user cannot guarantee how many views are captured, we will study training that keeps the model accurate with fewer views. The spoken result will be cautious: the glass will ask for a check by hand when it is not sure. We will also check the banknote watermark in a photograph of the note held against a light, by removing the effect of the lamp and comparing the watermark with a reference built from genuine notes of the same value.
- **Speech and control.** Offline text-to-speech (Piper and espeak-ng) with separate voices for Bangla and English segments; five push-buttons for mode, action, read and volume; every change of state is spoken.

Evaluation. Models will be trained on a training split, choices will be made on a validation split, and the test split will be used only for the final figures. Detection will be reported as precision, recall and mean average precision; reading as character and word error rate; the counterfeit check as accuracy and as the number of counterfeit notes wrongly passed. Latency, memory and temperature will be measured on the Raspberry Pi 5. Results that do not support our expectations will be reported as well.

7. Hardware and Software Requirements

Table 1: Hardware and estimated cost

Component	Role	Estimated cost (BDT)
Raspberry Pi 5 (4 to 8 GB) with cooler	Runs all models and the application	18,000 – 24,000
Raspberry Pi camera module	Captures the scene	1,500 – 2,500
Five push-buttons	Mode, action, read, volume up, volume down	100 – 200
Earphone or bone-conduction headset with amplifier	Speech output	800 – 1,500

Component	Role	Estimated cost (BDT)
MicroSD card (32 to 64 GB)	Operating system and model files	800 – 1,200
USB power bank	Portable power	1,500 – 3,000
3D-printed spectacle frame	Holds the camera and the board	1,500 – 2,500
Wiring, connectors, vibration motor	Supporting electronics	400 – 700
Total (approximate)		24,600 – 35,600

The costs are our estimates from local market prices and will be confirmed at purchase. Software will be open source: Python, OpenCV, PyTorch, Ultralytics YOLOv8, EasyOCR, ONNX Runtime, Piper and espeak-ng. Model training will be done on laptops with NVIDIA GPUs that the team already has.

8. Work Plan

Table 2: Work plan over one semester (16 weeks)

Weeks	Activity	Output
1 – 2	Literature review, requirement analysis, purchase of parts	Requirement list, parts in hand
3 – 5	Currency data pipeline and detector training	Trained Taka detector with test results
4 – 7	Text reading and object modes; speech pipeline	Working modes on a laptop with a webcam
6 – 9	Counterfeit study on the JaalTaka dataset	Trained classifier, safe-feedback rule
8 – 10	Frame design and 3D printing; button wiring	First wearable prototype
10 – 12	Integration and deployment on the Raspberry Pi 5	Glass running all modes from the buttons
12 – 14	Measurement on the Pi; trial sessions with users	Latency and accuracy figures, user feedback
14 – 16	Report writing and presentation	Thesis report and demonstration

9. Expected Outcomes

- A working prototype of an assistive glass that speaks Bangla, with text, object and currency modes.
- A labelled Taka detection dataset produced without manual annotation, and a detector trained on it.
- A study of camera-based counterfeit checking, with feedback wording that is safe for a blind user.

- Measured accuracy for every function and measured speed on the Raspberry Pi 5.
- Feedback from visually impaired users on ease of use, clarity of speech and usefulness.
- A thesis report and the source code, so that a later team can continue the work.

10. Risks and How We Will Handle Them

Table 3: Main risks

Risk	Plan
Models are too slow on the Raspberry Pi 5	Use small models, quantised ONNX exports, and run heavy tasks only on request
Bangla OCR is not accurate enough	Test pre- and post-processing and other engines; report the measured error honestly
A detector trained on synthetic images fails on real photographs	Use real photographs as backgrounds; test on photographs from independent sources
Few counterfeit notes are available	Use the public JaalTaka dataset; keep the spoken verdict cautious
Camera quality on the glass is lower than on a laptop	Test with the glass camera early; adjust resolution or change the camera
Hard to recruit visually impaired participants in time	Start contact early; report honestly how many sessions were completed

11. Ethical Considerations

Pictures will be processed on the device and will not be stored or sent anywhere in the offline modes. Participants in user trials will be told what the device does in spoken form, will take part only with their consent, and will be recorded under a number and not a name. Photographs of participants will be used only with their permission. Because a wrong counterfeit verdict could harm the user or another person, the glass will present its result as advice and will not accuse a note of being counterfeit.

12. Conclusion

SARA aims to give a blind person in Bangladesh a little more independence in three daily tasks: reading, knowing what is in front of them, and handling cash. The project combines existing open models with data and interaction designed for Bangla-speaking users, on hardware that is affordable and works without the internet. We ask for approval to carry out this work as our senior project under the supervision of Dr. Mohammad Shidujaman.

References

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